



Adyapan

School

C++ Programming



Duration - 3 months

**Industry
Certification**



Skill India Certified

**250+
Partner Companies**

Automate. Analyze. Achieve with C++

From code to creation - become industry-ready.

This immersive C++ Programming program takes you beyond foundational syntax into real-world software development. Learn object-oriented programming, data structures, memory management, and STL (Standard Template Library) through hands-on projects and guided problem-solving. With a strong focus on performance optimization and industry-relevant practices, you'll graduate with the ability to build efficient applications, write robust code, and tackle complex system-level challenges with confidence.

8

MODULES

30+

PROGRAM OFFERINGS

20,000+

STUDENTS

250+ PARTNERED COMPANIES

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course - a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders – defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS

Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

8 weeks. 8 modules. Infinite impact.

WEEK 1

C++ Fundamentals

- Discussion of Curriculum
- Introduction to Programming & C++ Setup (Compiler, IDE like VS Code / CodeBlocks)
- Structure of a C++ Program (main function, headers, namespaces)
- Variables, Data Types, Input/Output (cin, cout)
- Operators & Conditional Statements (if, switch)
- Loops (for, while, do-while)



WEEK 2

Core Programming Concepts

- Arrays & Strings in C++
- Pointers & Memory Basics (stack vs heap)
- Functions (pass by value vs reference)
- Recursion & Problem Solving
- File Handling (fstream)
- Exception Handling Basics



WEEK 3

Object-Oriented Programming (OOP)

- Classes & Objects
- Constructors, Destructors
- Encapsulation & Abstraction
- Inheritance (single, multiple, multilevel)
- Polymorphism (function overloading, virtual functions)
- Introduction to Header Files & Modular Programming



8 weeks. 8 modules. Infinite impact.

WEEK 4

STL & Advanced Data Handling

- Introduction to Standard Template Library (STL)
- Vectors, Lists, Deques
- Stacks, Queues, Priority Queues
- Maps & Sets
- Iterators & Algorithms (sort, find)



WEEK 5

Data Structures and Problem Solving

- Time & Space Complexity Basics
- Arrays, Linked Lists
- Stacks & Queues Implementation
- Trees (Binary Tree basics)
- Searching & Sorting Algorithms



WEEK 6

System-Level Programming

- Memory Management (new, delete)
- Dynamic Allocation & Smart Pointers (Intro)
- File Systems & Streams (advanced)
- Command-line arguments
- Multithreading Basics (<thread>)



8 weeks. 8 modules. Infinite impact.

WEEK 7

Application Development

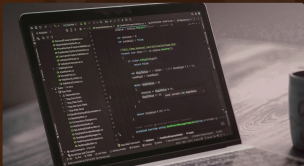
- Introduction to GUI (basic console UI design)
- Working with external libraries
- Database Connectivity
- API Integration basics (via libraries like libcurl)
- Project Architecture Design



WEEK 8

Full Stack Project

- Designing a complete C++ application
- Modular coding & file structuring
- Authentication logic (if applicable)
- Performance optimization techniques
- Debugging & Testing



WHO THIS IS FOR

This course is perfect for

Students & Career Switchers

Aspiring Software Developers

Non-Tech Learners Entering IT

Tech Entrepreneurs &
Freelancers

Analysts & Consultants

Automation & Scripting
Learners

CERTIFICATIONS



ALUMNI NETWORK

Our alumni work at world-class companies

Amazon

Adobe

Google

Autodesk

Microsoft

Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email us at support@adyapan.com

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